#10 多元复合函数求偏导题型总结

类型一 复合函数求一阶偏导

【例 1】（2007）设 $f(u,v)$ 为二元可微函数，$z=f\left(x^y,y^x\right)$，则 $\dfrac{\partial z}{\partial x}=$_____.

[解析] 由多元复合函数求导法则，有 $\dfrac{\partial z}{\partial x}=\dfrac{\partial f}{\partial u}\dfrac{\partial}{\partial x}\left(x^y\right)+\dfrac{\partial f}{\partial v}\dfrac{\partial}{\partial x}\left(y^x\right)=yx^{y-1}f_1'+\left(y^x\ln y\right)f_2'$

【例 2】（李永乐 660）设函数 $f(x,y)$ 可微，且 $f(1,2)=2,f_x'(1,2)=3,f_y'(1,2)=3,z(x)=f(x,f(x,2x))$，求 $z'(1)$.

[解析] $z'(x)=f_1'\left(x,f\left(x,2x\right)\right)+f_2'\left(x,f\left(x,2x\right)\right)\left[f_1'\left(x,2x\right)+2f_2'\left(x,2x\right)\right]$

由于 $f\left(1,f(1,2)\right)=f(1,2)$，$f_1'(1,2)=f_x'(1,2)=3,f_2'(1,2)=f_y'(1,2)=3$

$z'(1)=f_1'\left(1,f\left(1,2\right)\right)+f_2'\left(1,f\left(1,2\right)\right)\left[f_1'(1,2)+2f_2'(1,2)\right]=f_1'(1,2)+f_2'(1,2)\left[f_1'(1,2)+2f_2'(1,2)\right]$

$=3+3\times\left(3+2\times3\right)=30$.

[注] 求解过程中不宜采用记号 $f_x'(x,2f(x,2x))$，原因在于从这个记号无法区分开来究竟是表示函数 $z(x)=f(x,2f(x,2x))$ 对 x 的导数，还是表示 $f(x,2f(x,2x))$ 对第一个位置求偏导.

【例 3】设 $f(xy,x+y)=x^2+y^2+xy$.

（1）求二元函数 $z=f(x,y)$　　（2）求 $\dfrac{\partial z}{\partial x},\dfrac{\partial z}{\partial y}$

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[解析] $f(xy,x+y)=x^2+y^2+xy=-xy+(x+y)^2$，令 $u=xy,v=x+y$，有 $f(u,v)=-u+v^2$

故 $z=f(x,y)=-x+y^2$，容易求得 $\dfrac{\partial z}{\partial x}=-1,\dfrac{\partial z}{\partial y}=2y$.

类型二 复合函数求二阶偏导

【例 1】（2005）设函数 $u(x,y)=\varphi(x+y)+\varphi(x-y)+\int_{x-y}^{x+y}\psi(t)dt$，其中函数 φ 具有二阶导数，ψ 具有一阶导数，则必有(　).

A. $\dfrac{\partial^2 u}{\partial x^2}=-\dfrac{\partial^2 u}{\partial y^2}$　　B. $\dfrac{\partial^2 u}{\partial x^2}=\dfrac{\partial^2 u}{\partial y^2}$　　C. $\dfrac{\partial^2 u}{\partial x\partial y}=\dfrac{\partial^2 u}{\partial y^2}$　　D. $\dfrac{\partial^2 u}{\partial x\partial y}=\dfrac{\partial^2 u}{\partial x^2}$

[解析] 因为 $\dfrac{\partial u}{\partial x}=\varphi'(x+y)+\varphi'(x-y)+\psi(x+y)-\psi(x-y)$，$\dfrac{\partial u}{\partial y}=\varphi'(x+y)-\varphi'(x-y)+\psi(x+y)+\psi(x-y)$

于是 $\dfrac{\partial^2 u}{\partial x^2}=\varphi''(x+y)+\varphi''(x-y)+\psi'(x+y)-\psi'(x-y)$

$\dfrac{\partial^2 u}{\partial x\partial y}=\varphi''(x+y)-\varphi''(x-y)+\psi'(x+y)+\psi'(x-y)$

$\dfrac{\partial^2 u}{\partial y^2}=\varphi''(x+y)+\varphi''(x-y)+\psi'(x+y)-\psi'(x-y)$

可得 $\dfrac{\partial^2 u}{\partial x^2}=\dfrac{\partial^2 u}{\partial y^2}$，应选(B).

【例 2】（1992）设$z=f\left(\mathrm{e}^{x}\sin y,x^{2}+y^{2}\right)$,且$f$二阶连续可微,求$\dfrac{\partial^{2}z}{\partial x\partial y}$.

[解析]本题为复合函数求偏导.$z=f\left(\mathrm{e}^{x}\sin y,x^{2}+y^{2}\right)$两边对$x$求偏导得$\dfrac{\partial z}{\partial x}=\mathrm{e}^{x}\sin y\cdot f_{1}'+2xf_{2}'$，

$$\begin{aligned}\frac{\partial^{2}z}{\partial x\partial y}&=\frac{\partial}{\partial y}\left(\mathrm{e}^{x}\sin y\cdot f_{1}'\right)+\frac{\partial}{\partial y}\left(2xf_{2}'\right)\\&=\mathrm{e}^{x}\cos y\cdot f_{1}'+\mathrm{e}^{x}\sin y\cdot\left(\mathrm{e}^{x}\cos y\cdot f_{11}''+2yf_{12}''\right)+2x\left(\mathrm{e}^{x}\cos y\cdot f_{21}''+2yf_{22}''\right)\\&=\mathrm{e}^{x}\cos y\cdot f_{1}'+\mathrm{e}^{2x}\sin y\cos y\cdot f_{11}''+2\mathrm{e}^{x}(x\cos y+y\sin y)f_{12}''+4xyf_{22}''\end{aligned}$$

【例 3】（2005）设$f(u)$具有二阶连续导数,且$g(x,y)=f\left(\dfrac{y}{x}\right)+yf\left(\dfrac{x}{y}\right)$,求$x^{2}\dfrac{\partial^{2}g}{\partial x^{2}}-y^{2}\dfrac{\partial^{2}g}{\partial y^{2}}$.

[解析]由已知得，$\dfrac{\partial g}{\partial x}=-\dfrac{y}{x^{2}}f'\left(\dfrac{y}{x}\right)+f'\left(\dfrac{x}{y}\right)$，$\dfrac{\partial^{2}g}{\partial x^{2}}=\dfrac{2y}{x^{3}}f'\left(\dfrac{y}{x}\right)+\dfrac{y^{2}}{x^{4}}f''\left(\dfrac{y}{x}\right)+\dfrac{1}{y}f''\left(\dfrac{x}{y}\right)$，

$$\frac{\partial g}{\partial y}=\frac{1}{x}f'\left(\frac{y}{x}\right)+f\left(\frac{x}{y}\right)-\frac{x}{y}f'\left(\frac{x}{y}\right),$$

$$\frac{\partial^{2}g}{\partial y^{2}}=\frac{1}{x^{2}}f''\left(\frac{y}{x}\right)-\frac{x}{y^{2}}f'\left(\frac{x}{y}\right)+\frac{x}{y^{2}}f'\left(\frac{x}{y}\right)+\frac{x^{2}}{y^{3}}f''\left(\frac{x}{y}\right)=\frac{1}{x^{2}}f''\left(\frac{y}{x}\right)+\frac{x^{2}}{y^{3}}f''\left(\frac{x}{y}\right)$$

故$x^{2}\dfrac{\partial^{2}g}{\partial x^{2}}-y^{2}\dfrac{\partial^{2}g}{\partial y^{2}}=\dfrac{2y}{x}f'\left(\dfrac{y}{x}\right)+\dfrac{y^{2}}{x^{2}}f''\left(\dfrac{y}{x}\right)+\dfrac{x^{2}}{y}f''\left(\dfrac{x}{y}\right)-\dfrac{y^{2}}{x^{2}}f''\left(\dfrac{y}{x}\right)-\dfrac{x^{2}}{y}f''\left(\dfrac{x}{y}\right)=\dfrac{2y}{x}f'\left(\dfrac{y}{x}\right)$.

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